

Business Intelligence Lab Unit 1 |

Session 4

Experiment 4

OLAP Operations (Slice, Dice, Drill-up & Drill-down) using Microsoft Power BI

Aim

To perform Online Analytical Processing (OLAP) operations using Microsoft Power BI by creating multidimensional reports and applying Slice, Dice, Drill-down, and Drill-up operations on the Global Superstore dataset for analytical decision-making.

Tools / Software Required

1. Microsoft Power BI Desktop
2. Global Superstore Dataset.xlsx
3. Power Query Editor
4. DAX (Data Analysis Expressions)

Dataset Source:

Global Superstore Dataset (Kaggle)

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Theory

Online Analytical Processing (OLAP) is a Business Intelligence technique used for multidimensional analysis of business data. OLAP enables users to analyze information from different perspectives and perform complex analytical queries efficiently.

Common OLAP operations include:

Slice: Selecting a single dimension value from a dataset.

Dice: Selecting multiple dimensions and values simultaneously.

Drill-down: Moving from summarized data to detailed data.

Drill-up: Moving from detailed data to summarized data.

Power BI supports OLAP-style analysis through hierarchies, filters, slicers, visual interactions, and drill functionality.

Conduct - Part A: Load Dataset and Build Report

Step 1: Open Power BI Desktop.

Step 2: Open the Power BI model created in Experiment 3 or import the Global Superstore dataset.

Step 3: Verify relationships between Orders, Returns, and People tables.

Step 4: Create report visuals:

- Sales by Region
- Profit by Category
- Sales by Segment
- Sales Trend by Year

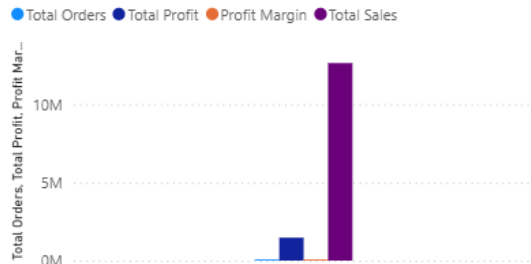


Conduct - Part B: Slice Operation

Step 5: Add a slicer visual using Region.

Step 6: Select a single Region such as West.

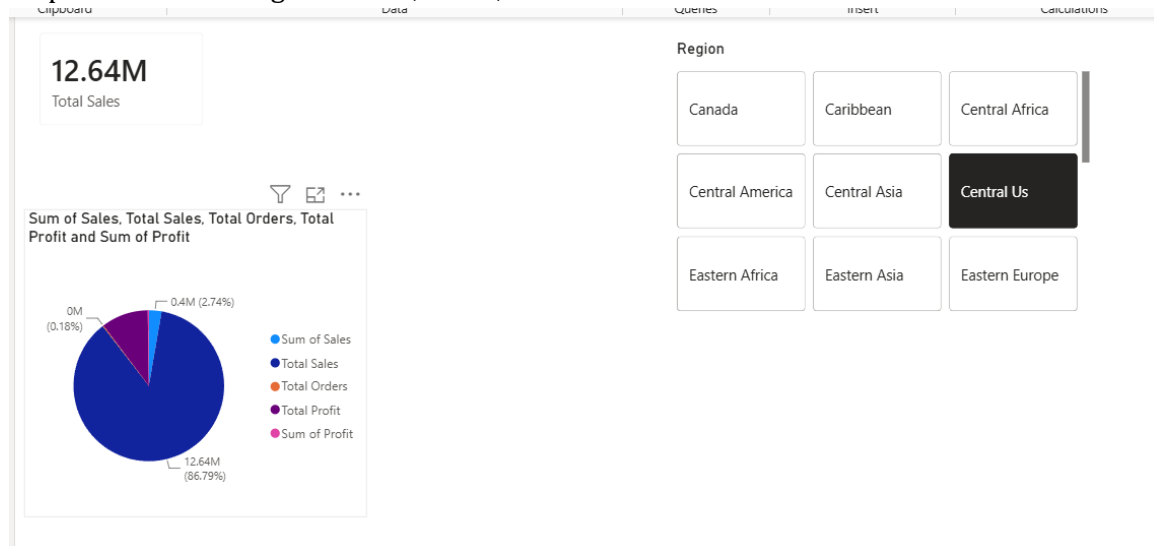
Total Orders, Total Profit, Profit Margin and Total Sales



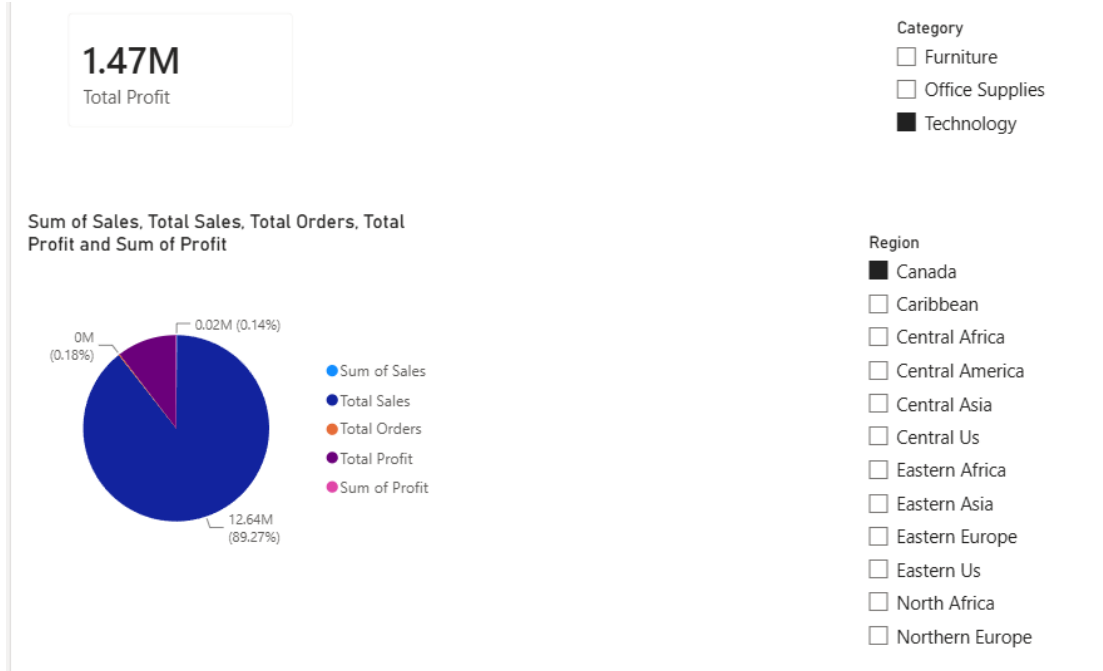
Total Profit by Region

- ☒ Canada
- ☐ Caribbean
- ☐ Central Africa
- ☐ Central America
- ☐ Central Asia
- ☐ Central Us
- ☐ Eastern Africa
- ☐ Eastern Asia
- ☐ Eastern Europe
- ☐ Eastern Us
- ☐ North Africa
- ☐ Northern Europe
- ☐ Oceania
- ☐ South America
- ☐ Southeastern Asia
- ☐ Southern Africa
- ☐ Southern Asia

Step 7: Observe changes in Sales, Profit, and Orders.



Step 8: Record observations and capture screenshot evidence.



Conduct - Part C: Dice Operation

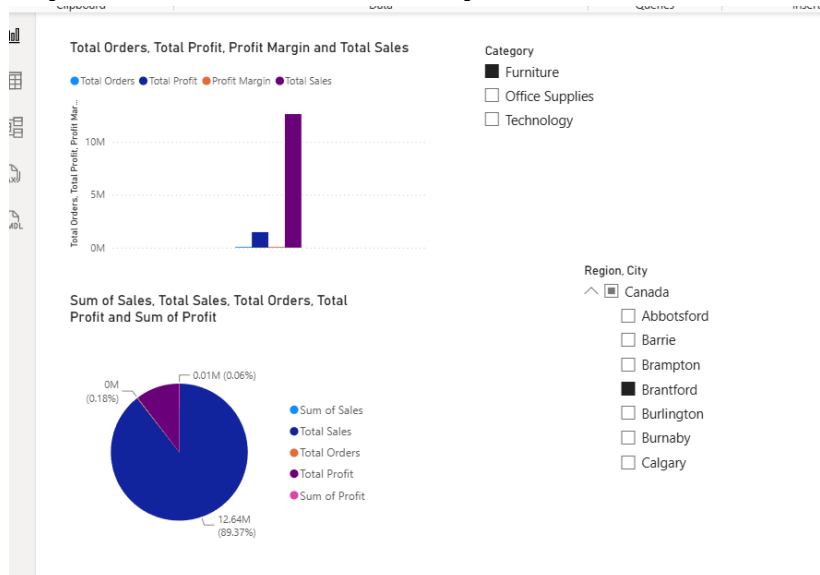
Step 9: Apply multiple filters simultaneously.

Example:

- Region = West
- Category = Technology
- Year = 2022

Step 10: Analyze filtered business metrics.

Step 11: Record observations and capture screenshots.



Conduct - Part D: Drill-down Operation

Step 12: Create a hierarchy using Region → State → City.

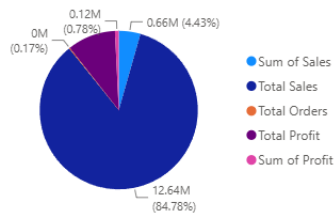
Step 13: Create a bar chart using the hierarchy.

Step 14: Use Drill-down mode to navigate from Region level to City level.

Step 15: Observe detailed sales patterns.

Step 16: Capture screenshots.

Sum of Sales, Total Sales, Total Orders, Total Profit and Sum of Profit



Country, State, Region, City

- ^ Afghanistan
 - ∨ Hirat
 - ^ Kabul
 - ∨ Southern Asia
 - ∨ Kandahar
 - ∨ Nangarhar
 - ∨ Albania
 - ∨ Algeria
 - ∨ Angola
 - ∨ Argentina

Conduct - Part E: Drill-up Operation

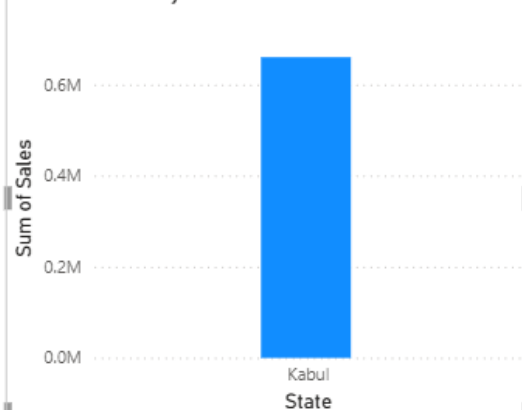
Step 17: Navigate back from City level to State level and then Region level.

- Total Profit
- Sum of Profit

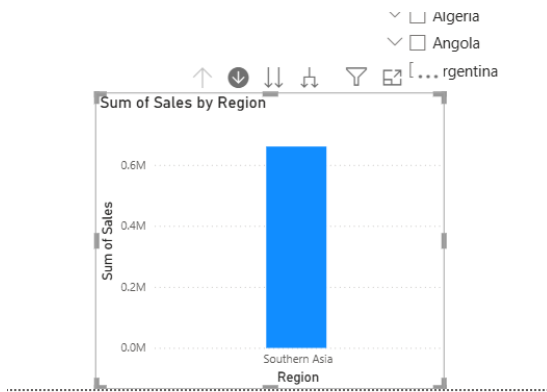
- ∨ Nangarhar
- ∨ Albania
- ∨ Algeria
- ∨ Angola
- [... Argentina

Drill up

Sum of Sales by State



Step 18: Compare aggregated values at each level.



Step 19: Record observations.

Drill-up from City to State and then Region showed aggregated sales values at each geographical level. The Region level provided an overall summary, while State and City levels provided more detailed information.

Step 20: Capture screenshots.

Conduct - Part F: OLAP Analysis

Step 21: Comparing results obtained from Slice, Dice, Drill-down, and Drill-up operations.

Operation	What I did	Result
Slice	Selected Western US region	Report showed data only for the selected region
Dice	Selected Western US + Technology + 2017	Report showed data for a specific combination of region, category and year
Drill-down	Went from Region → State → City	More detailed sales information was obtained
Drill-up	Went from City → State → Region	Detailed data was summarized into higher-level totals

Step 22: Creating a summary table documenting observations.

OLAP Operation	Observation	Business Use
Slice	Filters data based on one dimension, such as Region	Helps managers analyze performance of a particular region

OLAP Operation	Observation	Business Use
Dice	Filters data using multiple dimensions simultaneously	Helps identify performance for a specific region, category and year
Drill-down	Provides increasingly detailed information from Region to City	Helps identify specific states or cities with high/low sales
Drill-up	Aggregates detailed information from City to Region	Helps managers understand overall regional performance

Step 23: Explaining how each operation supports business decision-making.

1. Slice

Slice helps managers focus on one specific area.

Example: Selecting Western US allows the manager to analyze only Western US sales and profit.

2. Dice

Dice helps managers analyze a specific combination of conditions.

Example: Western US + Technology + 2017 can help determine how well technology products performed in that region during that year.

3. Drill-down

Drill-down helps managers find more detailed information.

Example: If a region has low sales, managers can drill from Region → State → City to identify where sales are low.

4. Drill-up

Drill-up provides a **higher-level summary**.

Example: After examining individual cities, managers can drill up to State and Region to understand the overall performance.

Result

OLAP operations including Slice, Dice, Drill-down, and Drill-up were successfully performed using Microsoft Power BI. Multidimensional analysis was carried out on the Global

Superstore dataset, enabling detailed and summarized views of business performance from multiple perspectives.

Learning Outcomes

- Explain OLAP concepts and multidimensional analysis. (CO3-BT3)
- Perform Slice and Dice operations using filters and slicers. (CO3-BT3)
- Create hierarchies and perform Drill-down and Drill-up analysis. (CO5-BT6)
- Interpret analytical insights generated through OLAP operations. (CO3-BT4)
- Design interactive analytical reports for business users. (CO5-BT6)